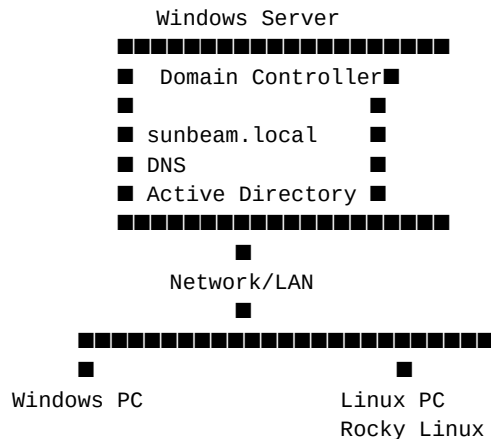


PowerShell, Active Directory & Linux Basics

Beginner-friendly study notes based on the uploaded notes.md

1. Big Picture

The material builds a Windows Server and Linux environment where the Windows Server can act as an Active Directory Domain Controller, while a Rocky Linux machine can later join the Windows domain.



2. PowerShell Basics

PowerShell is Microsoft's command-line shell and scripting environment. PowerShell commands are commonly called cmdlets and often follow a Verb-Noun pattern.

```
Get-ComputerInfo
Get-NetIPAddress
Get-NetAdapter
New-ADUser
Get-ADUser
Remove-NetIPAddress
Install-WindowsFeature
```

3. Get-Help

Use Get-Help to learn about a PowerShell command. It is one of the first commands to learn because it lets you look up syntax and examples instead of memorizing everything.

```
Get-Help Get-NetAdapter
Get-Help Get-NetIPAddress -Examples
```

4. Computer Information

Get-ComputerInfo displays information about the Windows computer, such as operating-system and system information.

```
Get-ComputerInfo
```

5. Network Adapters

Get-NetAdapter displays the network adapters/interfaces on the computer, such as Ethernet or Wi-Fi.

```
Get-NetAdapter
```

6. IP Address Information

Get-NetIPAddress displays configured IP addresses. You can restrict the result to a particular interface with -InterfaceAlias.

```
Get-NetIPAddress
Get-NetIPAddress -InterfaceAlias Ethernet
```

7. PowerShell Parameters

A parameter gives extra instructions to a command. A common pattern is: Command -Parameter Value.

```
Get-NetIPAddress -InterfaceAlias Ethernet
```

8. Changing an IPv4 Address

Remove-NetIPAddress removes an existing IP address from an interface. New-NetIPAddress assigns a new IP address. The uploaded notes use 192.168.1.100/24 as the example.

```
Remove-NetIPAddress -InterfaceAlias Ethernet -AddressFamily ipv4 -Confirm:$false
```

```
New-NetIPAddress -IPAddress 192.168.1.100 -PrefixLength 24 -AddressFamily ipv4 -InterfaceAlias Ethernet
```

-Confirm:\$false tells PowerShell not to ask for confirmation. Use this carefully with commands that modify or remove configuration.

9. Hostname

A hostname is the computer's name on the network. Rename-Computer changes it; hostname displays it.

```
Rename-Computer -NewName "DC01" -Restart
hostname
```

10. DNS Configuration

DNS is important in Active Directory because domain services rely on name resolution. The notes configure the Ethernet interface to use 192.168.1.100 as its DNS server.

```
Set-DnsClientServerAddress -InterfaceAlias Ethernet -ServerAddresses 192.168.1.100
Get-DnsClientServerAddress
```

11. Time Zone

Set-TimeZone changes the system timezone. Get-TimeZone displays the current timezone.

```
Set-TimeZone -Id "India Standard Time"
Get-TimeZone
```

12. Active Directory Domain Services

Active Directory Domain Services (AD DS) is the Windows Server directory service used in the lab to manage domain resources such as users, computers, groups, and organizational units.

13. Check AD DS Feature

Get-WindowsFeature checks the status of the AD Domain Services Windows Server feature.

```
Get-WindowsFeature Ad-Domain-Services
```

14. Install AD DS

Install-WindowsFeature installs the AD Domain Services feature and its management tools.

```
Install-WindowsFeature -Name Ad-Domain-Services -IncludeManagementTools
```

Installing AD DS is distinct from creating the domain. The next stage promotes/configures the server as a domain controller.

15. Create the Active Directory Forest

Install-ADDSForest is used in the notes to create the Active Directory forest and configure the domain name sunbeam.local. The command also enables DNS installation and specifies database, log, and SYSVOL paths.

```
Install-ADDSForest -DomainName "sunbeam.local" -DomainNetBiosName "SUNBEAM" -DomainMode "Win2025" -ForestMod
```

16. Verify Forest, Domain and Domain Controller

These commands retrieve information about the configured Active Directory environment.

```
Get-ADForest
Get-ADDomain
Get-ADDomainController
```

Get-ADForest → forest information

Get-ADDomain → domain information

Get-ADDomainController → domain-controller information

17. Organizational Units (OUs)

An Organizational Unit is a container used to organize Active Directory objects. The notes create an OU named Sunbeam Users.

```
New-ADOrganizationalUnit -Name "Sunbeam Users" -Path "DC=sunbeam,DC=local"
Get-ADOrganizationalUnit -Filter *
```

18. AD Objects: Computers, Users and Groups

The Get-ADComputer, Get-ADUser and Get-ADGroup commands retrieve corresponding Active Directory objects.

```
Get-ADComputer -Filter *
Get-ADUser -Filter *
Get-ADGroup -Filter *
```

19. Creating an AD User

New-ADUser creates a user account. The notes demonstrate a user named John Doe with the account name jdoe and a user principal name of jdoe@sunbeam.local.

```
New-ADUser -Name "John Doe" -GivenName "John" -Surname "Doe" -UserPrincipalName "jdoe@sunbeam.local" -SamAcc
```

20. Linux Networking

The Linux section uses Rocky/Linux-style commands. nmtui provides a text-based NetworkManager interface for configuring networking.

```
sudo nmtui
ip address show
ip a
```

21. Rebooting Linux

sudo reboot restarts the Linux machine. sudo runs the command with elevated privileges.

```
sudo reboot
```

22. Installing Domain-Join Packages

The notes install packages used for Linux integration with Active Directory, including realmd, SSSD, adcli, Samba tools and Kerberos workstation tools.

```
sudo dnf install -y realmd oddjob sssd oddjob-mkhomedir adcli samba-common-tools krb5-workstation
```

23. Discover the Active Directory Domain

realm discover checks whether the Linux machine can discover the specified domain.

```
sudo realm discover "sunbeam.local"
```

24. Join Linux to the Domain

realm join attempts to join the Linux machine to the Windows Active Directory domain using the specified account.

```
sudo realm join "sunbeam.local" -U Administrator
```

25. Complete Learning Sequence

Study the material in this order so that each topic builds on the previous one.

```
LEVEL 1 – PowerShell
Get-Help
Get-ComputerInfo
Get-NetAdapter
Get-NetIPAddress
hostname
```

```
LEVEL 2 – Windows Networking
Remove-NetIPAddress
New-NetIPAddress
Set-DnsClientServerAddress
Get-DnsClientServerAddress
Set-TimeZone
Get-TimeZone
Rename-Computer
```

```
LEVEL 3 – Active Directory
Get-WindowsFeature
Install-WindowsFeature
Install-ADDSForest
Get-ADForest
Get-ADDomain
Get-ADDomainController
```

```
LEVEL 4 – AD Management
New-ADOrganizationalUnit
Get-ADOrganizationalUnit
Get-ADComputer
Get-ADUser
```

```
Get-ADGroup
New-ADUser
```

LEVEL 5 – Linux Domain Integration

```
nmtui
ip a
dnf install
realm discover
realm join
```

26. Beginner Practice

Start by safely inspecting the system before making configuration changes. Run these commands in PowerShell:

```
Get-Help Get-NetAdapter
Get-ComputerInfo
Get-NetAdapter
Get-NetIPAddress
hostname
```

After you understand these inspection commands, move to networking configuration, DNS, Active Directory installation, AD object management, and finally Linux domain integration.

Quick Reference

PowerShell:

Get-Help	→ command help
Get-ComputerInfo	→ computer information
Get-NetAdapter	→ network adapters
Get-NetIPAddress	→ IP addresses
hostname	→ computer name
New-NetIPAddress	→ assign an IP
Rename-Computer	→ change computer name

Active Directory:

Get-WindowsFeature	→ check Windows features
Install-WindowsFeature	→ install AD DS
Install-ADDSForest	→ create/configure AD forest
Get-ADForest	→ forest information
Get-ADDomain	→ domain information
Get-ADDomainController	→ DC information
New-ADUser	→ create AD user

Linux:

nmtui	→ network configuration UI
ip a	→ IP/interface information
sudo reboot	→ restart Linux
dnf install	→ install packages
realm discover	→ discover AD domain
realm join	→ join AD domain